DTIT customer

Access to public internet / HowTo

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- Implicit (transparent proxy)

Abstract

Access to the public internet is possible through a GCP LZ web proxy for HTTP and HTTPS protocols.

The GCP LZ web proxy can be used as either explicit proxy or implicit (transparent) proxy.

Whenever possible, the explicit proxy should be used.

The implicit proxy is a kind of work-around for the clients which cannot be configured to use the explicit proxy.

The implicit proxy acts as a "man in the middle", it intercepts and terminates the HTTPS traffic from the clients and originates new connections to the destinations (e.g. web servers).

This approach results in a number of problems, the most important is that the clients cannot check the destination's certificate and should be configured to ignore certificate checking,

Also, the clients are not able to present their certificates to the destination.

Only the destinations (URL or IP) which were approved by security are whitelisted and can be accessed.

Every region (currently europe-west3 and europe-west4) in prod and dev environments has its own proxy.

In the same environment, one can access a proxy in any region from any region. This may be useful if a proxy in some region experiences a problem.

However, it is recommended to use the proxy in the same region to reduce unnecessary network overhead, whenever possible.

Explicit proxy

Prod environment

Region	URL	IP:PORT
Europe-west3	web-prd-ew3.gc.telekom.de:3128	100.64.3.10:3128

Region	URL	IP:PORT
Europe-west4	web-prd-ew4.gc.telekom.de:3128	100.64.4.10:3128

The proxy variables may be set like e.g.:

export http_proxy="http://web-prd-ew3.gc.telekom.de:3128"

export https_proxy="http://web-prd-ew3.gc.telekom.de:3128"

The proxy leverages GCP Cloud NAT to enable connectivity to public internet.

Currently, the following IP addresses are reserved for Cloud NAT:

(These are the source IP addresses of the outgoing IP packets, may be needed for firewall flows)

Europe-west3	Europe-west4
34.141.93.32	34.147.48.29
35.234.124.29	35.204.131.225
34.159.204.0	34.147.63.67

The current setup allows serving ~10000 clients per a region at the same time.

Later, new addresses may be added if needed.

Dev environment

Region	URL	IP:PORT
Europe-west3	web-dev-ew3.gc.telekom.de:3128	100.64.1.10:3128
Europe-west4	web-dev-ew4.gc.telekom.de:3128	100.64.2.10:3128

The proxy variables may be set like e.g.:

export http_proxy="http://web-dev-ew3.gc.telekom.de:3128"

export https_proxy="http://web-dev-ew3.gc.telekom.de:3128"

The proxy leverages GCP Cloud NAT to enable connectivity to public internet.

Currently, the following IP addresses are reserved for Cloud NAT:

(These are the source IP addresses of the outgoing IP packets, may be needed for firewall flows)

Europe-west3	Europe-west4
34.159.135.61	34.90.238.144
34.89.185.48	34.90.130.143
34.141.40.94	34.141.227.111

The current setup allows serving ~10000 clients per a region at the same time.

Later, new addresses may be added if needed.

Configuring system-wide proxy on a Linux VM

On most Linux systems, the system-wide proxy may be configured by putting a file into the /etc/profile.d/ directory, e-g. /etc/profile.d/proxy.sh with the following content:

> *cat /etc/profile.d/proxy.sh*
> *export http_proxy="http://web-prd-ew3.gc.telekom.de:3128"*
> *export https_proxy="http://web-prd-ew3.gc.telekom.de:3128"*

Implicit (transparent proxy)

The clients which cannot use the explicit proxy should configure the implicit proxy as a default gateway.

This is done by simply assigning a network tag to the VM where the client runs.

Note: this is possible to run clients (software programs) able to use explicit proxy together with the clients which are not able to use explicit proxy on the same VM.

To use the implicit proxy, the VMs must be configured with the following network tags:

	Prod	Dev
europe-west3	use-impl-proxy-europe-west3	use-impl-proxy-europe-west3
europe-west4	use-impl-proxy-europe-west4	use-impl-proxy-europe-west4

Like in case of explicit proxy, one can access a proxy in any region from any region in the same environment. This may be useful if a proxy in some region experiences a problem.

However, it is recommended to use the proxy in the same region to reduce unnecessary network overhead, whenever possible.

The implicit proxy uses the same instance Cloud NAT, so the source IP addresses for the outgoing traffic are also the same.

Check the organization policies inherited in your project

While managing/creating GCP resources, like assigning an external IP address to a instance, you may get a constraint violation error message like: "Constraint constraints/compute.vmExternalIpAccess violated for project".

In this case, you should first make sure that the constraint is not defined on your Project or on a Folder in which your Project lives.

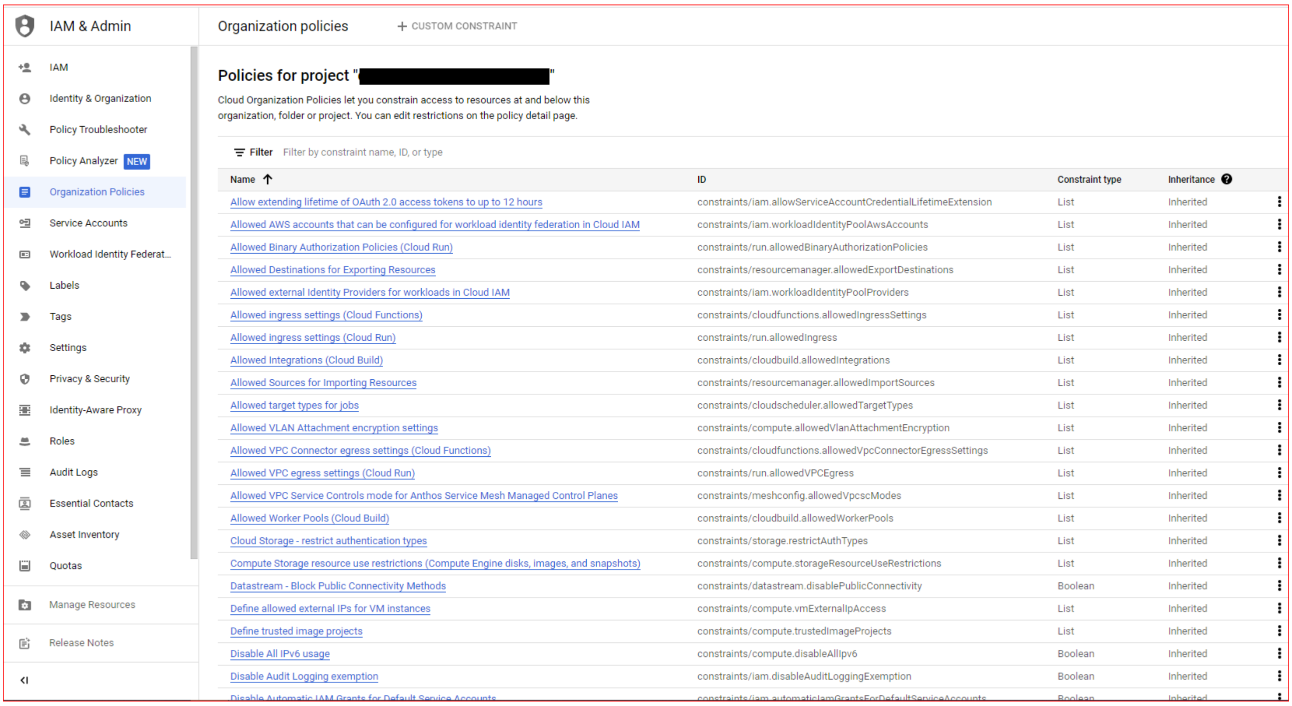
- If it is, you should contact the admins of the Project/Folder about the policy,
- If not, it may be inherited from an organization policy.

You can check the organization policies applied to your project using the following link: **https://console.cloud.google.com/iam-admin/orgpolicies/list?project=${PROJECT_NAME}**, where you must replace **${PROJECT_NAME}** with a valid project name.

If you are not able to open this page, it might be because you lack some of the required permissions. All needed permissions to access this page:

- orgpolicy.constraints.list
- orgpolicy.policies.list
- resourcemanager.projects.get

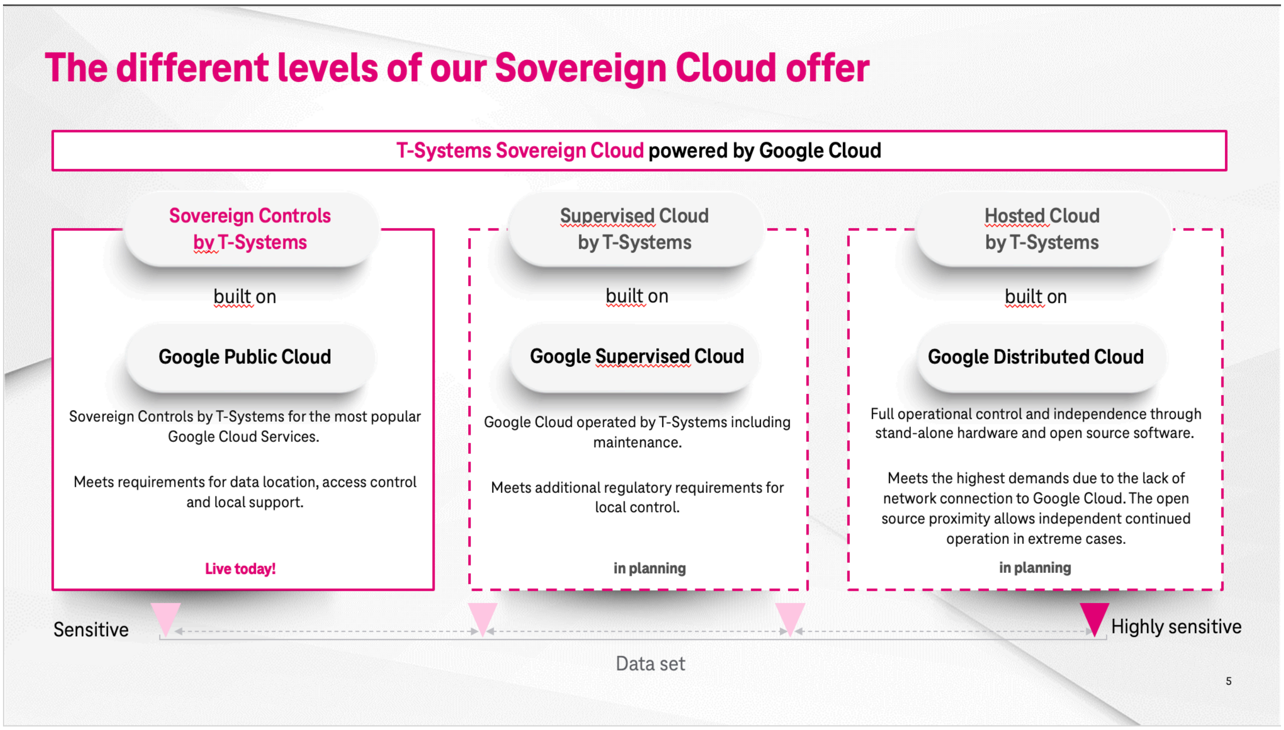
If you have the above permissions, you should see a similar page shown below. Here you can find all the organization policies that apply to your project.



Name	ID	Constraint type	Inheritance
Allow extending lifetime of OAuth 2.0 access tokens to up to 12 hours	constraints/iam.allowServiceAccountCredentialLifetimeExtension	List	Inherited
Allowed AWS accounts that can be configured for workload identity federation in Cloud IAM	constraints/iam.workloadIdentityPoolAwsAccounts	List	Inherited
Allowed Binary Authorization Policies (Cloud Run)	constraints/run.allowedBinaryAuthorizationPolicies	List	Inherited
Allowed Destinations for Exporting Resources	constraints/resourcemanager.allowedExportDestinations	List	Inherited
Allowed external Identity Providers for workloads in Cloud IAM	constraints/iam.workloadIdentityPoolProviders	List	Inherited
Allowed ingress settings (Cloud Functions)	constraints/cloudfunctions.allowedIngressSettings	List	Inherited
Allowed ingress settings (Cloud Run)	constraints/run.allowedIngress	List	Inherited
Allowed Integrations (Cloud Build)	constraints/cloudbuild.allowedIntegrations	List	Inherited
Allowed Sources for Importing Resources	constraints/resourcemanager.allowedImportSources	List	Inherited
Allowed target types for jobs	constraints/cloudscheduler.allowedTargetTypes	List	Inherited
Allowed VLAN Attachment encryption settings	constraints/compute.allowedVlanAttachmentEncryption	List	Inherited
Allowed VPC Connector egress settings (Cloud Functions)	constraints/cloudfunctions.allowedVpcConnectorEgressSettings	List	Inherited
Allowed VPC egress settings (Cloud Run)	constraints/run.allowedVPCEgress	List	Inherited
Allowed VPC Service Controls mode for Anthos Service Mesh Managed Control Planes	constraints/meshconfig.allowedVpcscModes	List	Inherited
Allowed Worker Pools (Cloud Build)	constraints/cloudbuild.allowedWorkerPools	List	Inherited
Cloud Storage - restrict authentication types	constraints/storage.restrictAuthTypes	List	Inherited
Compute Storage resource use restrictions (Compute Engine disks, images, and snapshots)	constraints/compute.storageResourceUseRestrictions	List	Inherited
Datastream - Block Public Connectivity Methods	constraints/datastream.disablePublicConnectivity	Boolean	Inherited
Define allowed external IPs for VM instances	constraints/compute.vmExternalIpAccess	List	Inherited
Define trusted image projects	constraints/compute.trustedImageProjects	List	Inherited
Disable All IPv6 usage	constraints/compute.disableAllIpv6	Boolean	Inherited
Disable Audit Logging exemption	constraints/iam.disableAuditLoggingExemption	Boolean	Inherited
Disable Automatic IAM Grants for Default Service Accounts	constraints/iam.automaticIamGrantsForDefaultServiceAccounts	Boolean	Inherited

More info on Organizational Policies

Data sovereignty product overview



How to add additional users to my project

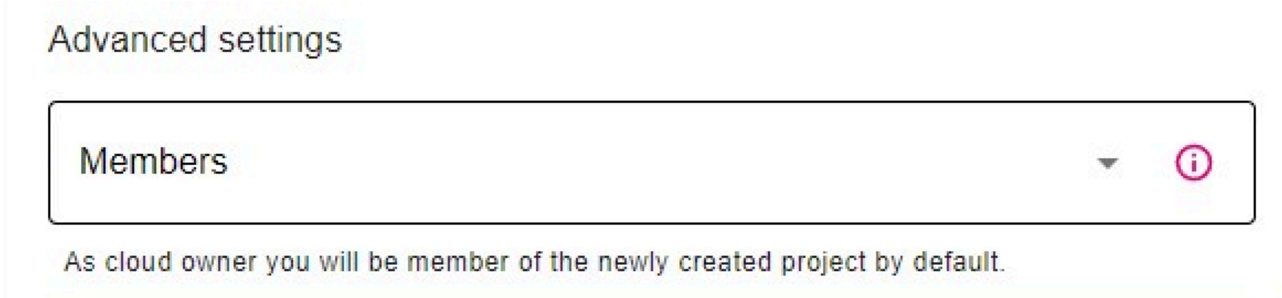
- Add additional user by initial project creation

- Add users once the project is created

Important Note: only users with GCP user can be added to the projects. To order Google Cloud user see the Onboarding Page

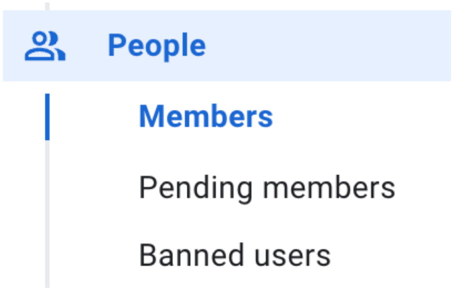
Add additional user by initial project creation

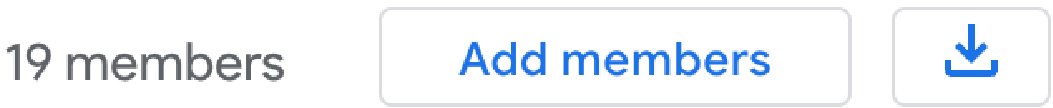
To add additional users to my project there is a possibility by the project creation in Cloud Shephard



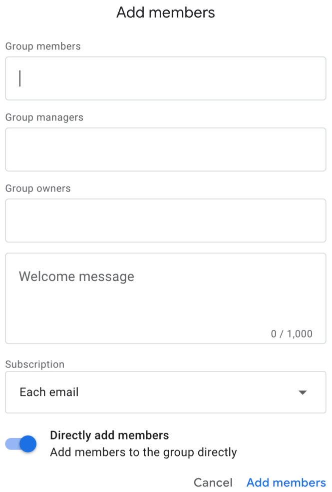
Add users once the project is created

1. Navigate to the website: https://groups.google.com/
2. Select the Group where you want to edit, e.g. select the "members" group of your project, the groups name is something like "**mgt-de0360-<dev|prd>-<project>-<suffix>-members**".
3. On the left site select **Members** in People section, and click on Add Members on the Top of the page





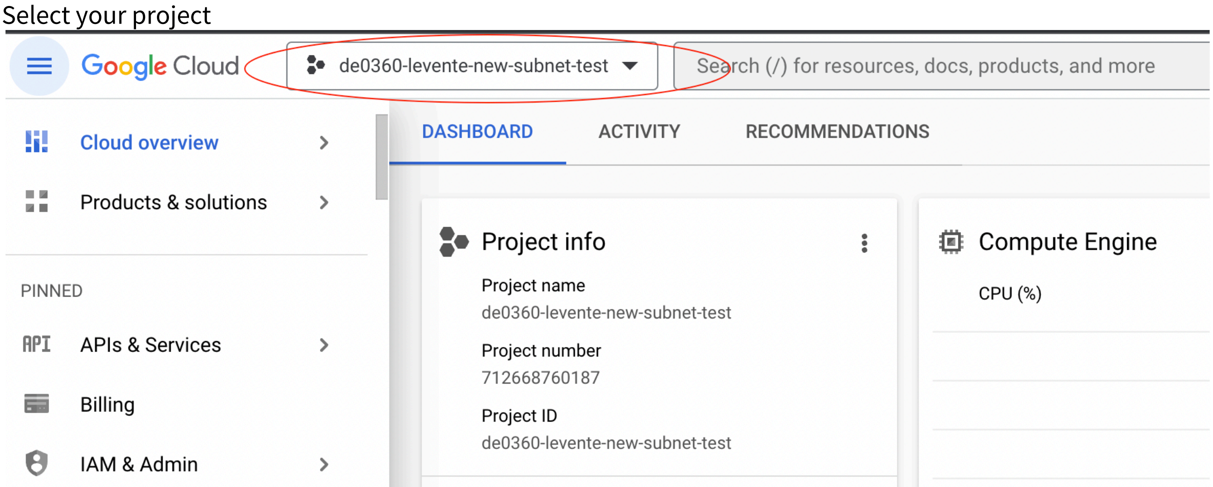
4. In the pop-up window add the user eMail to the permission level. You can add Member, Manager, and Owner. Only Manager and Owner can add further Member to the Group



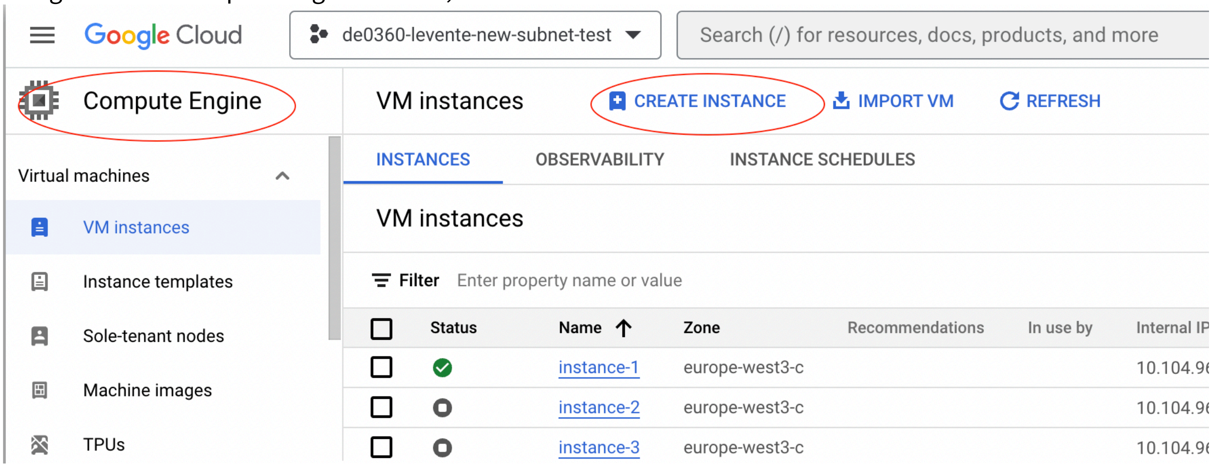
5. You can optionally specify a Welcome message, and then click on Add members Button

How to create a Compute Engine (VM)

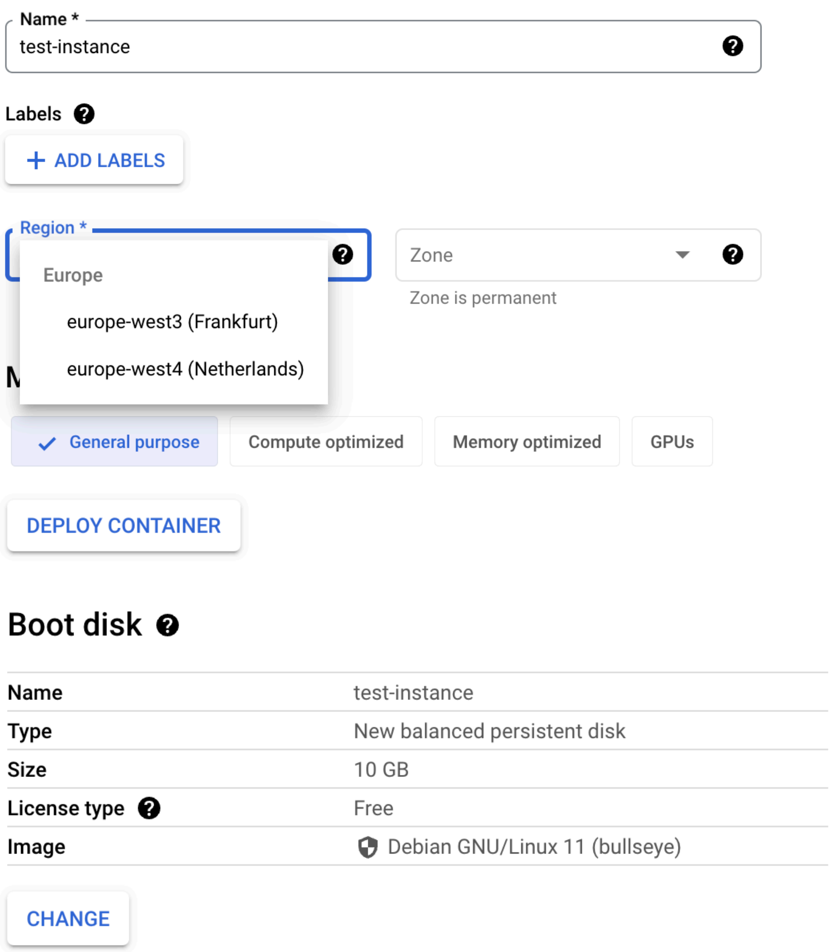
1. Log in to Google Cloud Console: https://console.cloud.google.com/
2. Select your project



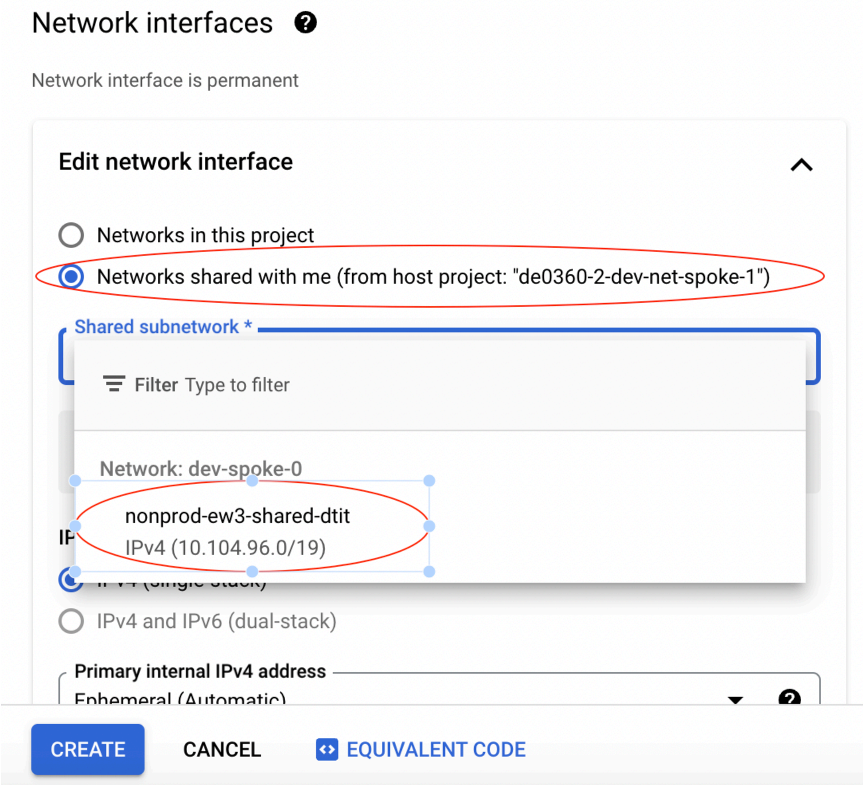
3. Navigate to the Compute Engine section, and click on **Create Instance**



4. Add a **Name** for the instance, and chose the **Region** and **Zone** (only Frankfurt, and Netherland allowed), set the **instance size**, and the **OS** you want to have.



5. Open the **Advanced options** to set up Network. Scroll to Network Interfaces, and select the shared network



6. You can add additional settings (e.g. persistent disk) by this section. Then click on Create to Create VM

Create VM with CLI (Cloud Shell)

You can create VM-s also via CloudShell. For the previous example here you can find the respective Code:

Create VM

```
gcloud compute instances create test-instance \
  --project=de0360-levente-new-subnet-test \
  --zone=europe-west3-c --machine-type=e2-medium \
  --network-interface=stack-type=IPV4_ONLY,subnet=projects/de0360-2-dev-net-spoke-1/regions/europe-west3/
subnetworks/nonprod-ew3-shared-dtit,no-address \
  --metadata=enable-oslogin=true \
  --maintenance-policy=MIGRATE \
  --provisioning-model=STANDARD \
  --service-account=712668760187-compute@developer.gserviceaccount.com \
  --scopes=https://www.googleapis.com/auth/devstorage.read_only,https://www.googleapis.com/auth/
logging.write,https://www.googleapis.com/auth/monitoring.write,https://www.googleapis.com/auth/
servicecontrol,https://www.googleapis.com/auth/service.management.readonly,https://www.googleapis.com/auth/
trace.append \
  --create-disk=auto-delete=yes,boot=yes,device-name=test-instance,image=projects/debian-cloud/global/
images/debian-11-bullseye-v20230509,mode=rw,size=10,type=projects/de0360-levente-new-subnet-test/zones/us-
central1-a/diskTypes/pd-balanced \
  --no-shielded-secure-boot \
  --shielded-vtpm \
```

```
--shielded-integrity-monitoring \
--labels=goog-ec-src=vm_add-gcloud \
--reservation-affinity=any
```

Note: as of now communication between VMs in your project only works on special ports and protocols (443, 80, 22, rdp, icmp). If you need more open ports, you can request Firewall Rules in our Support Queue. Best way for ordering a Firewall rule is that you specify source and target Tags you want connection between (e.g. "tier/application" tag to "tier/database" tag). Then you must tag your source and target VM instances in order for the Firewall rule to take effect (see Tags for Firewalls documentation).

Private Service Connect

A Private Service Connect Endpoint is deployed for the customer DE0360. This Endpoint is using the network prd-landing-0. The subnet 10.104.5.0/24 is reserved for PSC.

The endpoint with name '**dtag**' is configured with IP **10.104.5.5**. This IP is routed in CNDTAG and via our Interconnect to the Google Cloud.

You can use all Google API according to the following naming scheme:

`<SERVICE>-dtag.p.googleapis.com`

Add the PSC names as A record to your DNS or in your local '/etc/hosts'.

Some examples:

Service	Default API	Our Private Service Connect Endpoint	IP
Compute Engine API	compute.googleapis.com	compute-dtag.p.googleapis.com	10.104.5.5
Cloud Pub/Sub API	pubsub.googleapis.com	pubsub-dtag.p.googleapis.com	10.104.5.5
Cloud Storage API	storage.googleapis.com	storage-dtag.p.googleapis.com	10.104.5.5
Kubernetes Engine API	container.googleapis.com	container-dtag.p.googleapis.com	10.104.5.5
BigQuery API	bigquery.googleapis.com	bigquery-dtag.p.googleapis.com	10.104.5.5

To use the PSC Endpoint you have also to rewrite the default endpoint:

```
gcloud config set api_endpoint_overrides/<SERVICE> <ENDPOINT_URL>
```

Example for Storage API:

```
gcloud config set api_endpoint_overrides/storage https://storage-dtag.p.googleapis.com/storage/v1/
```

Get more help and endpoint names: https://cloud.google.com/sdk/gcloud/reference/config/set → Search for 'api_endpoint_overrides'.

Tests

Test DNS:

```
dig a storage-dtag.p.googleapis.com
```

Answer must be 10.104.5.5!

☐

Test routing and access:

```
curl -Ivk "https://10.104.5.5/generate_204"
```

Access must be without Proxy!

☐

Test TLS and access:

```
curl -Iv "https://storage-dtag.p.googleapis.com/generate_204"
```

☐

Access must be without Proxy!

☐

Tips

Get the default entpoint URL:

Set the API endpoint to default and use it with 'debug' verbosity .

```
gcloud config unset api_endpoint_overrides/storage
gcloud storage buckets list --verbosity=debug
```

```
[...]
DEBUG: Starting new HTTPS connection (1): storage.googleapis.com:443
DEBUG: https://storage.googleapis.com:443 "GET /storage/v1/b?alt=json&maxResults=1000&project=de0360-2-prd-net-landing-1&projection=full
```

```
HTTP/1.1" 200 32
[...]
```

Get the API endpoint (Serivce name)...

Documentation

Private Service Connect endpoint documentation: https://cloud.google.com/vpc/docs/configure-private-service-connect-apis#using-endpoints

Setting up a pipeline

Preparing your project for the pipeline

Forced by the fact that service tokens arent allowed to create for deployments, we have to go the way of workload identity federation (WIF).

The steps in the following blueprint project will help you to generate the service account and the workload identity pool strings in google cloud, which you need in your pipeline to deploy to your project

https://gitlab.devops.telekom.de/ccoe/teams/gcloud-landing-zone/blueprints/gitlab-ci

After it you can setup your pipeline in gitlab.

This are yaml codeblocks from an example which worked to set up the pipeline.

.gitlab-ci.yml

```
# Include your test templates and files here
#
#

# Adjust what you need
variables:
  NO_PROXY: "NO_PROXY=localhost,127.0.0.1"
  TF_ROOT: ${CI_PROJECT_DIR}
  GOOGLE_CREDENTIALS: .gcp_temp_cred.json
  TF_GC_docker_image: mtr.devops.telekom.de/gclz/cloud-foundation-cicd:1
```

```
  # here we store the terrafom state in the gitlab
  # if you want an other solution you arent bound to this
  TF_ADDRESS: ${CI_API_V4_URL}/projects/${CI_PROJECT_ID}/terraform/state/${CI_PROJECT_NAME}
  TF_USERNAME: "${TF_VAR_my_stateuser_name}" # gitlab user name given acces to store TF state file
  TF_PASSWORD: "${TF_VAR_my_stateuser_token}" # gitlab user password given acces to store TF state file

  # here include your credentials from the WIF setup
  TF_MYWIF: "projects/1234567890/locations/global/workloadIdentityPools/gitlab-devops-telekom-de/providers/
gitlab-devops-telekom-de-oidc"
  TF_MYWIFSA: "automation@de0360-my-google-projekt.iam.gserviceaccount.com"

cache:
  key: $CI_PROJECT_NAME
  paths:
    - ${TF_ROOT}/.terraform
    - ${TF_ROOT}/.terraform.lock.hcl
    - ${GOOGLE_CREDENTIALS}

before_script:
  - cd ${TF_ROOT}

# Some script functions for the WIF setup
.google_WIF_auth:
  script:
  - 'echo ${CI_JOB_JWT_V2} > .ci_job_jwt_file;gcloud iam workload-identity-pools create-cred-config $
{TF_MYWIF} --service-account=${TF_MYWIFSA} --output-file=.gcp_temp_cred.json --credential-source-
file=.ci_job_jwt_file'

stages:
  - prepare
  - test
  - validate
  - build
  - deploy


init:
  stage: prepare
  image: ${TF_GC_docker_image}
  script:
    - !reference [.google_WIF_auth, script]
    - |
      terraform init \
        -backend-config="address=${TF_ADDRESS}" \
        -backend-config="username=${TF_USERNAME}" \
        -backend-config="password=${TF_PASSWORD}"


validate:
  stage: validate
  image: ${TF_GC_docker_image}
  script:
    - terraform validate

plan:
  stage: build
```

```
  image: ${TF_GC_docker_image}
  script:
    - !reference [.google_WIF_auth, script]
    - terraform plan -out plan.json
  artifacts:
    name: plan
    paths:
      - ${TF_ROOT}/plan.cache
    reports:
      terraform: ${TF_ROOT}/plan.json


apply:
  stage: deploy
  image: ${TF_GC_docker_image}
  script:
    - !reference [.google_WIF_auth, script]
    - terraform apply -auto-approve
  dependencies:
    - plan
  when: manual
  only:
    - main


destroy:
  stage: deploy
  image: ${TF_GC_docker_image}
  script:
    - !reference [.google_WIF_auth, script]
    - terraform destroy
  dependencies:
    - plan
  when: manual
  only:
    - main
```

backend.tf

```
terraform {
  backend "http" {
  }
}
```

some lines out of the vars.tf

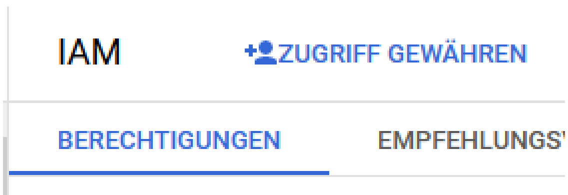
```
variable "zone" {
  type        = string
  nullable    = false
  description = "Google Cloud zone [a,b,c,d,e] in region"
```

```
  default     = "b"
}

variable "region" {
  type        = string
  nullable    = false
  description = "Google Cloud region"
  default     = "europe-west3"
}

variable "project" {
  type        = string
  nullable    = false
  description = "Google Cloud project ID."
  default     = "my-google-project"
}
```

Adjusting the rights of the automation user

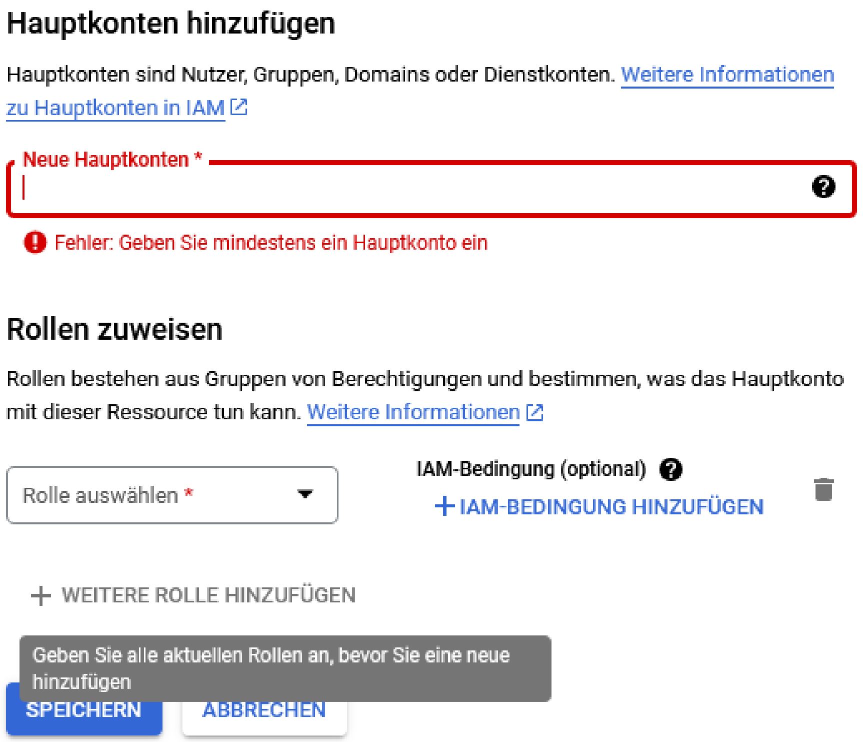
Out of the box the automation user (automation@<projectid>iam.gserviceaccount.com) hasn't got any rights. You have to add the nescessary rights you need manually. This can be done in your IAM area.

Choose "Zugriff gewähren" or even the englisch phrase.



Add the user in the corresponding field and choose the nescessary roles.

For example "Storage Admin" for creating buckets, or "Network admin" for adding IPs would be nescessary. For getting the right roles for used tasks please consut the google documentation.


Hauptkonten hinzufügen
Hauptkonten sind Nutzer, Gruppen, Domains oder Dienstkonten. Weitere Informationen zu Hauptkonten in IAM
Neue Hauptkonten *
Fehler: Geben Sie mindestens ein Hauptkonto ein
Rollen zuweisen
Rollen bestehen aus Gruppen von Berechtigungen und bestimmen, was das Hauptkonto mit dieser Ressource tun kann. Weitere Informationen
Rolle auswählen *
IAM-Bedingung (optional)
IAM-BEDINGUNG HINZUFÜGEN
WEITERE ROLLE HINZUFÜGEN
Geben Sie alle aktuellen Rollen an, bevor Sie eine neue hinzufügen
SPEICHERN
ABBRECHEN